

DANA 4840

Classification II

Cluster Analysis

Part 2

Use cases of cluster analysis

Cluster analysis is used in many fields.
For example:

- City administrators all over the world are using cluster analysis to segment houses within a city (eg for property taxes purpose)
- A large company may collect data from its 20,000 employees to decide (a) groups where training is needed and (b) groups deserving of increments and promotions

Similarity vs dissimilarity measurements

It is important to know the difference between a similarity measurement and a dissimilarity measurement.

- When a measurement increases when 2 objects (people) are alike (similar), we call that measurement a similarity measurement (eg Jaccard, dice coefficients etc)
- On the other hand, when a measurement decreases when 2 objects are alike, we call that measurement a dissimilarity measurement (eg Euclidean distance).

Methods to calculate similarity and dissimilarity measurements

(a) For binary (categorical) data use:

- i) simple matching
- ii) Jaccard
- iii) dice

These are similarity measurements

(b) For quantitative data, we can use:

- i) Euclidean distance
- ii) Manhattan distance
- iii) Correlation-based distances

Methods to calculate similarity and dissimilarity measurements

(c) For mixed data (combination of quantitative and categorical data in a dataset), use **Gower distance**.

In R, you can obtain gower distance using `daisy()` command.

Teach students: how to convert a similarity measurement to a dissimilarity measurement and why we need to do that.

Simple matching, Jaccard and dice coefficient calculations

- Simple matching $\frac{a+d}{a+b+c+d}$

- Jaccard $\frac{a}{a+b+c}$

- Dice $\frac{2a}{2a+b+c}$

Simple matching, Jaccard and
dice coefficient calculations

Example: